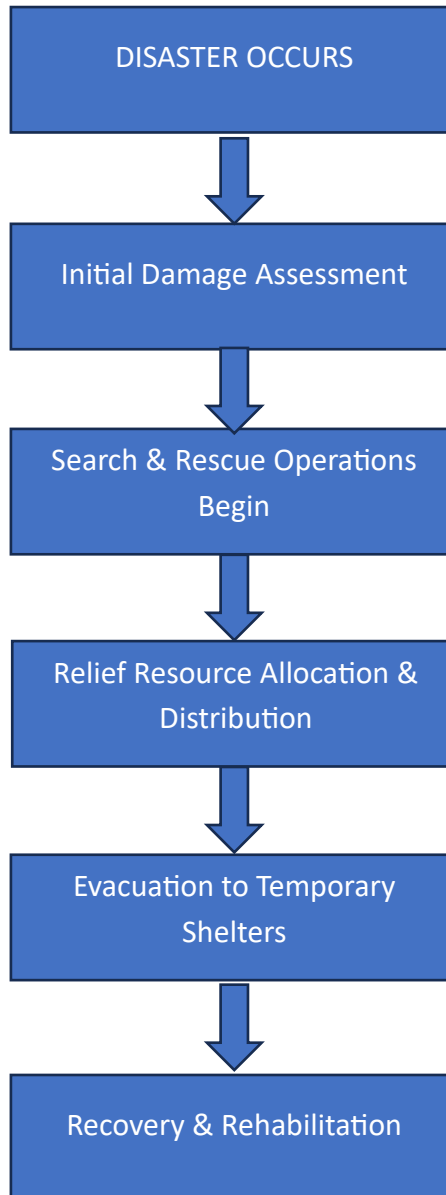


AETHER

AI-Powered Disaster Response and Relief Management System

Emergency Phase Workflow:



1. Pain Points in Current Disaster Response

These should be **high-level operational pain points**, not technology-specific ones. Every pain point should naturally map to one of your modules.

1. Delayed Survivor Localization

- Rescue teams spend valuable time searching large disaster zones with limited information.
- Existing search methods are manpower-intensive and often rely on manual inspection.

- Poor visibility, debris, smoke, and damaged infrastructure delay victim identification.
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2. High False Positives During Search

- Individual sensing methods (thermal, radio, visual) are unreliable when used independently.
 - Abandoned devices, animals, fire hotspots, or environmental interference can trigger incorrect detections.
 - Rescue teams waste time verifying false alarms.
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3. Lack of Real-Time Situational Awareness

- Emergency operation centers often receive fragmented and delayed information.
 - Field teams, shelters, and logistics units operate using different communication channels.
 - Decision-makers lack a unified operational picture.
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4. Poor Visibility of Relief Inventory

- Food, water, medicines, and rescue equipment are frequently tracked manually.
 - Inventory updates are delayed or inaccurate.
 - Authorities cannot quickly identify shortages or available resources.
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5. Inefficient Shelter Allocation

- Shelter occupancy information is rarely updated in real time.
 - Evacuees may be directed to shelters that are already full.
 - Rescue teams spend additional time identifying alternative shelters.
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6. Communication Disruptions

- Cellular and internet infrastructure may become unavailable after disasters.
 - Field teams struggle to synchronize information across multiple locations.
 - Critical updates are delayed.
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7. Manual Coordination Across Multiple Teams

- Rescue, logistics, medical, and shelter management teams often work using separate systems.
 - Information must be manually relayed between departments.
 - Lack of integration increases operational complexity.
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8. Lack of Data-Driven Decision Support

- Existing workflows depend heavily on human judgment.
 - There is limited use of real-time analytics to prioritize rescue operations or allocate resources.
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2. SMART Problem Statement

To reduce the time required for disaster response teams to identify probable survivors, manage critical relief resources, and allocate evacuees to suitable shelters by developing an integrated, real-time decision-support platform that combines multi-sensor UAV detection, connected inventory management, and live shelter intelligence over resilient communication networks.

SMART Breakdown

Specific

- Develop a unified disaster-response dashboard integrating survivor detection, inventory monitoring, and shelter management.

Measurable

- Reduce survivor search time.
- Improve inventory visibility.
- Minimize shelter overcrowding.
- Reduce manual coordination between response teams.

Achievable

- Uses commercially available technologies including UAVs, BLE, thermal sensing, digital asset tags, and Cisco networking solutions.

Relevant

- Addresses critical operational challenges during the first hours of disaster response, where rapid decision-making has the greatest impact.

Time-Bound

- Designed to provide continuous real-time situational awareness throughout emergency response operations, particularly during the critical first 24–72 hours following a disaster.

3. Proposed Solution

4.1 Solution Overview

Introduction

To address the operational challenges identified in current disaster response systems, we propose **AETHER (AI-Powered Disaster Response and Relief Management System)**, an integrated decision-support platform that assists emergency responders throughout the critical response phase of a disaster.

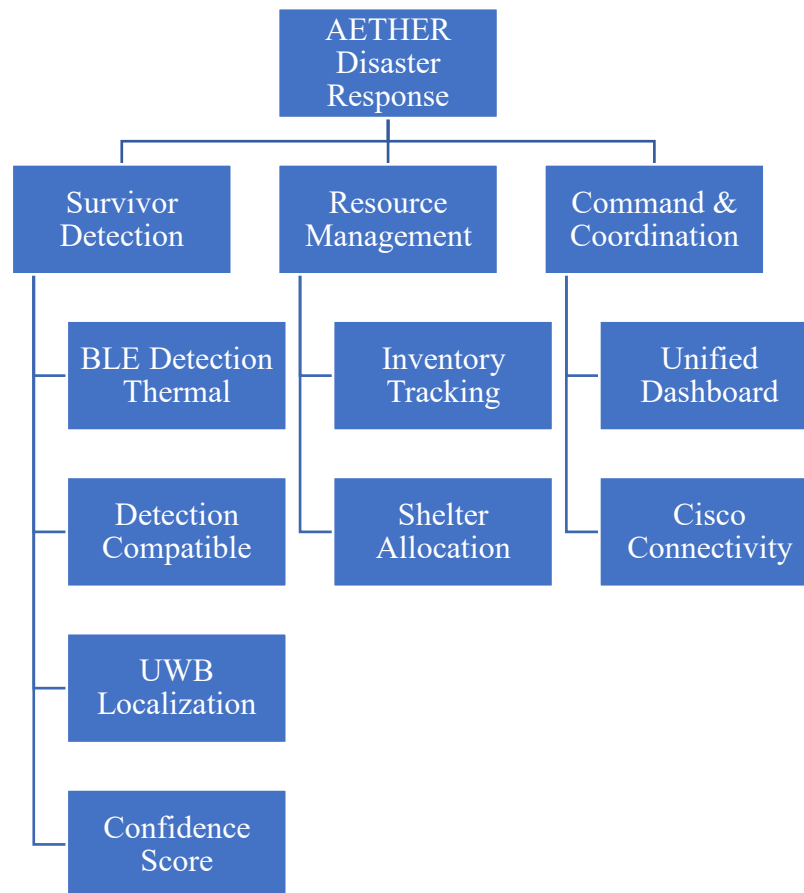
Rather than replacing existing rescue operations, AETHER enhances them by providing **real-time situational awareness, intelligent survivor localization, automated relief inventory tracking, and dynamic shelter management** through a unified command dashboard.

The platform follows a simple operational philosophy:

Detect → Verify → Equip → Evacuate

By integrating sensing technologies, connected infrastructure, and centralized decision support, AETHER enables responders to make faster, data-driven decisions while reducing manual coordination and operational delays.

4.2 Solution Workflow



4.3 Core System Components

AETHER consists of three primary operational modules:

1. *Multi-Sensor Survivor Detection*

The **Multi-Sensor Survivor Detection** module enables rapid identification of probable survivors using UAVs equipped with **Bluetooth Low Energy (BLE)**, **Thermal Imaging**, and **compatible Ultra-Wideband (UWB)** technologies. Data from these sensors is fused through a **confidence scoring mechanism** to reduce false positives and prioritize high-probability survivor locations, enabling rescue teams to respond faster and more efficiently.

2. *Smart Relief Inventory Management*

The **Smart Relief Inventory Management** module automates the tracking of essential relief supplies using **RFID/Digital tags**. Inventory levels are updated in real time whenever resources are issued or received, while low-stock alerts and centralized resource

visibility help authorities efficiently monitor and distribute supplies across warehouses and shelters.

3. Intelligent Shelter Management

The **Intelligent Shelter Management** module provides real-time information on shelter occupancy, capacity, and availability. By combining live shelter data with **GPS-based recommendations**, the system directs evacuees to the nearest suitable shelter, reducing overcrowding and improving evacuation planning during emergencies.

4.4 How AETHER Solves Each Pain Point

Pain Point	AETHER Solution
Delayed survivor localization	UAV-based multi-sensor detection with confidence scoring
High false positives	BLE + Thermal + UWB sensor fusion with confidence evaluation
Lack of situational awareness	Unified real-time command dashboard
Manual inventory tracking	RFID/Digital tag-based inventory management
Shelter overcrowding	Live shelter occupancy monitoring and recommendations
Communication disruptions	Cisco resilient mesh networking with offline synchronization
Poor coordination	Centralized operational dashboard
Slow decision making	Data-driven analytics and real-time updates

4.5 Why This Solution?

AETHER adopts a **multi-layered sensing strategy**, where each component compensates for the limitations of the others.

- **Bluetooth Low Energy (BLE)** provides broad device discovery due to its widespread availability in smartphones and wearables. ⁱ
- **Thermal imaging** offers visual confirmation of potential human presence, especially in low-visibility conditions. ⁱⁱ

- **Ultra-Wideband (UWB)**, when supported by compatible devices or dedicated tags, improves localization accuracy through precise ranging. ⁱⁱⁱ
- **Confidence scoring** combines evidence from multiple sensors to reduce false positives and prioritize rescue efforts. ^{iv}
- **RFID-based inventory management** automates resource tracking, minimizing manual errors and improving logistics. ^v
- **Cisco's resilient networking infrastructure** enables continuous communication and synchronization even in degraded connectivity environments.

This layered approach provides a more reliable and operationally practical solution than relying on any single technology alone.

5. Technical Architecture

5.1 Overall Technology Stack

Layer	Technology	Purpose
UI/Frontend	React.js + Tailwind CSS	Interactive web dashboard
UI Design	Figma	Dashboard wireframing and UI mockups
Backend	FastAPI (Python)	API development and sensor data processing
Database	PostgreSQL / Firebase	Store survivor, inventory and shelter data
Real-time Communication	WebSockets	Live dashboard updates
Drone	DJI / PX4 / ArduPilot	UAV deployment
BLE Scanner	Bluetooth 5.x Module	Detect nearby BLE devices
UWB	Decawave DWM1001 / Qorvo	Precise ranging
Thermal Camera	FLIR Lepton	Human heat detection
Inventory	RFID Reader	Resource tracking

Layer	Technology	Purpose
Maps	Google Maps API / Leaflet	Survivor & shelter visualization
Networking	Cisco Meraki Mesh	Communication infrastructure
Analytics	Splunk	Monitoring and dashboards

5.2 Frontend Architecture

The user interface of AETHER is developed using **React.js** to provide a responsive and modular dashboard capable of displaying live disaster information. Individual dashboard components such as survivor locations, inventory status, shelter availability, and operational analytics are implemented as reusable React components, while **Tailwind CSS** is used for rapid UI development and consistent styling. The initial user interface and workflow were designed in **Figma**, allowing the team to prototype the dashboard before implementation.

5.3 Backend Architecture

The backend is implemented using **FastAPI**, which exposes REST APIs and WebSocket endpoints for communication between drones, IoT devices, and the command dashboard. Incoming sensor data is validated, processed, and stored before being forwarded to the frontend. FastAPI was selected due to its asynchronous architecture, high performance, and native integration with Python-based AI and data processing libraries.

5.4 Sensor Data Acquisition

The survivor detection module receives data from three primary sensing technologies mounted on the UAV.

The **Bluetooth Low Energy (BLE)** scanner continuously scans for nearby smartphones and wearable devices, recording parameters such as device identifier, received signal strength (RSSI), timestamp, and drone GPS coordinates.

The **thermal camera** captures infrared images that are processed to identify potential human heat signatures and their approximate location within the scanned region.

When compatible devices or dedicated emergency tags are available, the **Ultra-Wideband (UWB)** module performs precise ranging measurements to improve localization accuracy.

5.5 Data Processing Pipeline

Raw sensor data collected by the UAV is transmitted to the backend server through the communication network. Each sensor produces different types of data, requiring preprocessing before analysis.

For BLE, the server extracts parameters such as RSSI values, signal stability, and detection frequency. Thermal images are processed using computer vision algorithms to identify probable human signatures, while UWB measurements provide distance estimates from compatible devices. These independent observations are then combined through the confidence scoring engine to estimate the probability of a survivor being present at a specific location.

5.6 Database Design

The backend maintains separate collections for survivor detections, inventory records, shelter information, and operational logs.

Each survivor record stores its GPS coordinates, confidence score, detected sensor sources, timestamp, and rescue status. Inventory records maintain stock levels and transaction history, while shelter records continuously update occupancy, capacity, and operational status. This structured storage enables efficient querying and real-time synchronization across all dashboard modules.

5.7 Communication Architecture

AETHER uses **WebSockets** to deliver real-time updates from the backend to the command dashboard without requiring manual page refreshes. Sensor events, inventory changes, and shelter updates are immediately pushed to connected clients, ensuring that responders always operate using the latest available information.

5.8 Cisco Integration

Communication between field units is supported through **Cisco Meraki Mesh Networking**, providing resilient connectivity in environments where conventional infrastructure may be disrupted. Cisco Spaces enables location-aware services and asset visibility, while Splunk collects operational logs and analytics to assist command centers in monitoring ongoing rescue operations.

5.9 Deployment

The frontend dashboard is deployed using **Vercel**, enabling rapid updates and continuous deployment directly from the GitHub repository. Backend services can be deployed using Docker containers on cloud platforms such as AWS, Azure, or Google Cloud, allowing the system to scale based on operational requirements.

6. End Remarks & Conclusion

6.1 End Remarks

Disaster response is one of the most time-critical operations, where even small delays in locating survivors or coordinating relief efforts can have significant consequences. Existing disaster management systems often rely on fragmented workflows, manual coordination, and disconnected information sources, making it difficult for emergency responders to obtain a unified operational picture during the critical response phase.

AETHER addresses these challenges by integrating survivor detection, relief inventory management, and shelter coordination into a single decision-support platform. Rather than replacing existing rescue operations, the platform enhances them by providing responders with timely, actionable information that enables faster and more informed decision-making. Its modular architecture also allows additional technologies and future capabilities to be incorporated without requiring major changes to the overall system.

6.2 Conclusion

AETHER demonstrates how modern sensing technologies, connected infrastructure, and intelligent software can work together to improve disaster response. By combining multi-sensor UAV-assisted survivor detection, automated relief inventory tracking, and real-time shelter management within a unified dashboard, the platform reduces operational complexity while improving coordination among emergency responders.

Although the current implementation focuses on the disaster response phase, the proposed architecture provides a scalable foundation for future enhancements such as AI-driven evacuation routing, predictive resource allocation, satellite-based communication, and digital twin integration. As disasters continue to increase in frequency and intensity, platforms such as AETHER have the potential to support emergency agencies in delivering faster, more coordinated, and more effective humanitarian assistance.

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